

Signal Distortion Analyzer Based on C2000

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Abstract

In the measurement and test of modern wireless communication, the accuracy of many parameters involves the problem of distortion measurement, especially in the process of signal generation, transmission and reception, the signal distortion must be measured and analyzed accurately. Signal distortion analyzer is a special instrument for measuring distortion, which is widely used in the production and detection of electronic products. This paper introduces a new signal distortion analyzer based on harmonic analysis method. The system consists of signal conditioning circuit, signal processing core board, display control panel and mobile APP(Application) display. The signal processing MCU1(Microcontroller Unit) and display control MCU2 all use C2000 series. The system takes the periodic signal with harmonic component from the signal source as the input signal, amplifies the amplitude and level of the signal conditioning circuit, and turns it into a signal matching analog-to-digital conversion ability of MCU1. Quantitative sampling using in-chip ADC(Analog to Digital Converter), FFT(Fast Fourier Transform) operation and periodic scan of MCU1, based on peak detection and windowed interpolation algorithm, achieves amplitude-frequency detection in non-integer cycle, and then calculates the normalized amplitude of the four harmonics and the base wave, as well as the signal distortion. MCU1 transmits signal characteristic data to MCU2 through serial port. After the waveform is recovered by MCU2, MCU1 displays the signal on screen and transmits it to APP display on mobile phone via Bluetooth. The test results show that the system has good stability and high measurement accuracy.

1 Introduction

The traditional distortion measurement instrument generally adopts the fundamental wave suppression method. The fundamental wave is suppressed by the filter, and the distortion is calculated from the total effective value of voltage and the effective value of harmonic voltage after suppressing the fundamental wave. The input signals are repetitive and can be designed to match the statistical and spectral characteristics of any given modulation format^[1]. However, this method will increase the system complexity and the measurement accuracy is not high.

The system which our team studies uses digital signal processing algorithm for harmonic analysis. The measured signal is sampled and the amplitude of each subharmonic and the base wave is calculated by FFT to calculate the total harmonic distortion. This system has the advantages of fast measurement speed and high measurement accuracy.

This system is developed with MCU of C2000 series TMS320F28335^[2]. Its ADC sampling rate can reach 12.5 Msps. It has a DMA and hardware multiplier for faster Fourier transformation.

Therefore, select C2000 series TMSF28335 with higher main frequency. It not only has the high-speed operation and signal processing capabilities of general DSP (Digital Signal Processing) chips, but also integrates rich

peripherals in the chip like MCU. It is suitable for high-performance digital control systems.

The system consists of signal conditioning module, TMS320F28335 core board and TMS320F2812 core board of C2000 series, screen display and mobile app display. The overall scheme is shown in **Figure 1**.

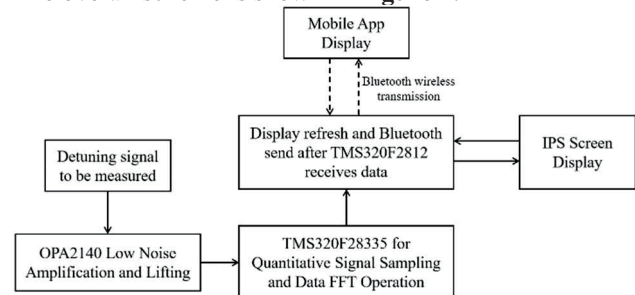


Figure 1 Design scheme of overall system architecture

The OPA2140 low noise operational amplifier can be converted into a signal matching the ADC voltage conversion range after four times amplification and voltage elevation of the measured detuning signal. The system is built with two MCUs because screen refresh and Bluetooth transmission consume more CPU (Central Processing Unit) resources and time, which affects ADC collection and FFT calculation. MCU1 specializes in ADC acquisition and FFT calculation. MCU1 calculates the amplitudes of the fundamental wave and each harmonic wave of the signal to be measured. MCU2 obtains MCU1 calculated signal

characteristic data for display driver. The screen rate of the system refreshes quickly and automatically measures the whole process without manual channel switching or sampling rate.

2 Theoretical analysis

2.1 Distortion Measurement Analysis

When the input of the amplifier is a sinusoidal signal, the nonlinear distortion of the amplifier is represented by the harmonic component distortion in the output signal. THD (Total Harmonic Distortion) is usually used to quantitatively analyze the degree of non-linear distortion of amplifiers^[3]. This system is limited to 5th harmonic. Harmonic distortion formula definition:

$$THD_0 = \frac{\sqrt{U_{o2}^2 + U_{o3}^2 + U_{o4}^2 + U_{o5}^2}}{U_{o1}} \quad (1)$$

When the THD_x value is measured by the harmonic analysis method, the ADC sampling and FFT processing of the measured signal are performed to obtain the amplitudes of the base wave and the sub-harmonics so as to bring them into the formula calculation.

2.2 Fast Fourier Transform

Fourier transform energy converts complex problems of signals in time domain to analysis in frequency domain^[4].

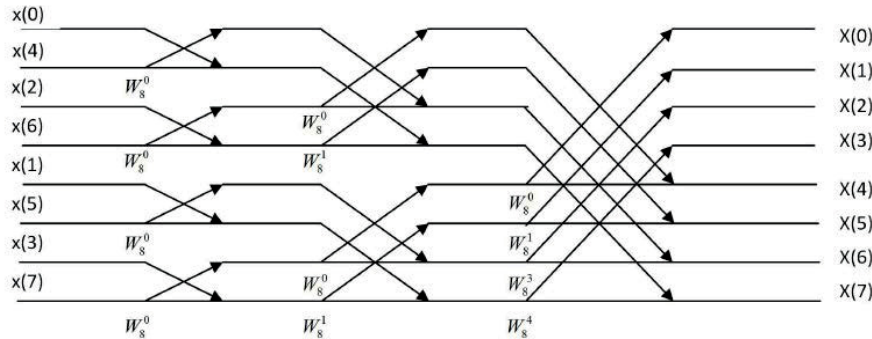


Figure 2 Eight Point Butterfly Operation

Fourier transforms are performed on each set of data. The Fourier transform results of the original time series are obtained by combining the Fourier transform results according to certain rules. The more points you make, the more operations you will save. Considering the storage capacity of the MCU controlled by this system, the overall accuracy is 4096 points.

Signals are continuous in frequency domain. Digital signal processing can only store and process discrete signals of limited length. It is not possible to directly perform continuous integral operations, so the signal needs to be discretized. In **Figure 3**, when we analyze the signal to be measured, we only intercept part of it. Even if the fundamental frequency of the signal obtained by FFT is calculated first to adjust the sampling rate, the period

Any function can be decomposed into the sum of an infinite number of sinusoidal signals of different frequencies. The signal components decomposed by Fourier transform analysis can be used to observe the characteristics of each frequency point in the frequency domain. This provides a theoretical basis for the harmonic study of this system. FFT (Fast Fourier Transform) uses the periodicity and symmetry of the W_N^{kn} factor to reduce the number of sine and cosine multiplications by decomposing a long time series step by step until the last set of data has only two points. Fourier transforms are performed on each set of data and the results are combined according to certain rules to obtain the Fourier transforms of the original time series. The MCU used by the system has a hardware multiplier to further increase the speed.

If there is a finite length discrete signal $f(n)$, the FFT expression is:

$$F(k) = \sum_{n=0}^{N-1} f(n)W_N^{kn} \quad (2)$$

If $N = 2^m$, $F(k)$ can be calculated by dividing it into odd and even sequences. Both sequences are $\frac{N}{2}$ periodic signals, and W_N^{kn} is the rotation factor. The formula is:

$$F_h = \sum_{k=2n} f_k W_N^{kh} + \sum_{k=2n+1} f_k W_N^{kh} \quad (3)$$

cannot be guaranteed to be an integer number. If you are in a non-integer cycle, you need to add windows to reduce leakage.

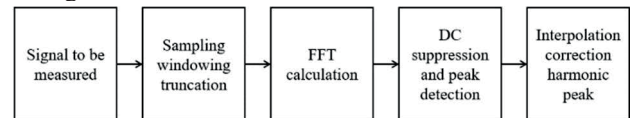


Figure 3 FFT signal processing

2.3 System Error Analysis

When the analog digital signal converter digitizes the analog signal, the maximum error is $\pm \frac{1}{2}q$. The q is the lowest significant bit. The quantization error is a uniform

distribution with a mean value of zero within $\pm \frac{1}{2}LSB$.

The standard deviation is $0.29LSB$. The 12 bit ADC used increases the digitization error by $\frac{0.29}{4096} = 0.007\%$ with little influence.

There are two ways to improve the quantized SNR (Signal-to-Noise Ratio). First, increase the quantized word length by about 6dB for each additional word length. The second is to increase the sampling frequency, which increases the signal-to-noise ratio by about 3dB each time the sampling frequency is doubled. The MCU used has determined that the number of bits in the ADC cannot be increased anymore, so the PSD of the quantized noise can be reduced by adjusting the sampling rate^[5]. Formula analysis is:

$$SNR = 6.02N + 1.76 + 10 \log \left(\frac{F_s}{2B} \right) \quad (4)$$

N is the quantized number of bits. F_s is the sampling rate. B is the bandwidth. With the increase of F_s , SNR can be improved. When the sampling frequency is increased, it will become oversampled. If the number of sampling points is insufficient, it will lead to a complete cycle that will not be collected by a single sample. Therefore, this design adjusts the sampling rate by real-time detecting the base frequency of the signal to be measured, eliminates other frequency band signals, and ensures noise error reduction. However, oversampling simply spreads the noise over a larger bandwidth. Invalid signals need to be filtered in different frequency domains after FFT operation to achieve pseudo-filter effect. It is still not possible to filter out noise from useful signals.

One of the main features of ADC acquisition and FFT operation is that they can operate on small signals. In order to avoid external signal interference, a high signal-to-noise ratio operational amplifier is needed to amplify the signal. Gain flatness bandwidth should be large enough, otherwise different amplification multiples for different frequency signals will result in greater measurement error of signal distortion. The maximum amplitude of the amplified signal needs to match the ADC sampling range, otherwise the accuracy will be affected or even the in-chip ADC will be damaged.

2.4 Operational Speed Analysis

In order to reduce the operation time, the code segment of TMS320F28xx MCU is located in FLASH. Placing required data and variables such as FFT operations within SRAM can greatly increase the speed of operations. Serial measurement data show that the time to complete single signal acquisition and calculate each harmonic normalized value and THD value is less than 1.5s.

3 Hardware and algorithm design

3.1 Signal conditioning circuit design

Signal conditioning circuit achieves signal amplitude amplification and DC offset. As shown in **Figure 4**, it uses a dual-amp OPA2140 with a low noise of 5.1nV/Hz and a GBW of 11 MHz. The test circuit has the same gain from 1 kHz to 800kHz. If the feedback resistance and input resistance are adjusted so that the AC signal is amplified by a factor of 4, there will be no harmonic component test error due to the difference in the suppression coefficients of different frequencies. By adjusting the potentiometer, the signal can be superimposed with a DC offset of 1.5 VPP. After the input of 30mVPP~600mVPP tuning, the signal is adjusted to the ADC-compliant collection voltage range. **Figure 4** is drawn by hardware circuit drawing software Altium Designer 21.

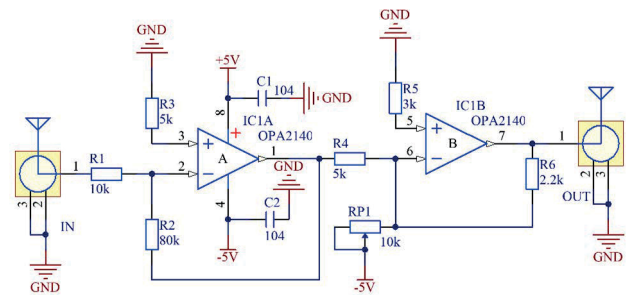


Figure 4 Signal conditioning circuit design

3.2 Software process

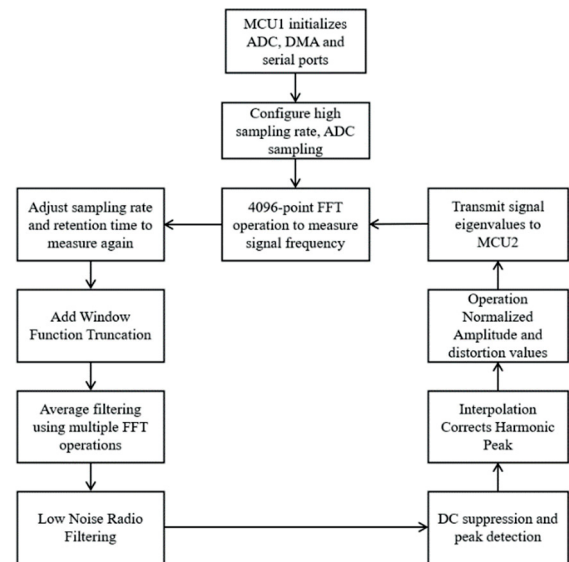


Figure 5 MCU1 Data Acquisition Processing and Transfer Flowchart

When the system is in each signal test process, the signal frequency is first determined by FFT operation of the signal samples. After adjusting the corresponding ADC configuration, signal quantization sampling and distortion calculation are performed. As shown in **Figure 5**, it is a software flowchart of data collection, processing and

transmission. It takes less than one second to complete two 4096 ADC sampling and FFT calculation through experimental tests. Data sampling and processing rates meet system design requirements.

As shown in **Figure 6**, the system uses TMS320F2812 MCU as the main controller. The data received by the serial port is checked for flag bits, and the data is converted after being confirmed to be correct. If the THD range is judged to be up to standard, the data is displayed on the screen and sent to the mobile APP via Bluetooth. After Bluetooth is connected to the APP end of the mobile phone, the data flags are constantly detected. If the flag bit correctly represents the correct data, refresh the data on the mobile page after successful reception.

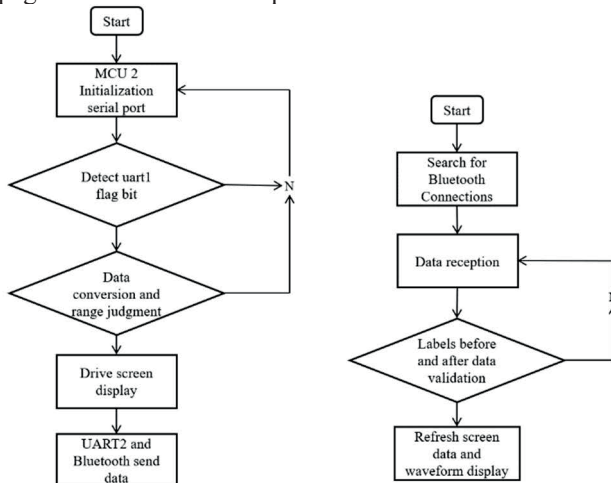


Figure 6 MCU2 Data Processing and Mobile APP Software Process

3.3 FFT Algorithm

As the main frequency domain analysis method, Fourier transform has a good effect on the harmonic detection and analysis in the signal. Converting time-domain signals to frequency domain clearly reflects the amplitude and phase of harmonics. However, there are some problems in the system, such as the truncation of the periodic signal, the spectrum leakage caused by frequency fluctuation, and the fence effect. These problems can lead to large errors in calculating the frequency and amplitude of each subharmonic.

To solve the problem, it is necessary to use windowing to obtain the actual spectrum of the signal. To solve the problems such as spectrum leakage, a window function that can increase the width of the main lobe, have a smaller peak level of the side lobe and a significant attenuation speed of the side lobe needs to be selected. Hanning window is used to truncate the ADC converted signal sequence before FFT operation. The idea of dividing FFT butterfly operations into two reduces the amount of data operations. It improves the computational efficiency by N times and saves time for performing FFT analysis multiple times^[6]. We analyze the data to design a data filter function. Filtering the results collected by FFT in a short time and many times can increase the correction accuracy, and make the system have

good anti-jamming characteristics and higher calculation accuracy.

3.4 DMA Transmission

The system realizes the transmission and storage of signal ADC conversion data through DMA (Direct Memory Access) technology^[7]. DMA saves the CPU resource overhead of MCU in the process of data sample collection. It realizes real-time acquisition and real-time transmission, and further reduces the waiting time of the system before FFT calculation. Through the analysis of the data, we design a digital filter.

4 Conclusion

For the system test, we mainly test the accuracy of THD and the system test time. Firstly, the signal source is adjusted to sine wave, the four harmonics detected are normalized to zero. The value of THD_x calculation result is zero. Set up the mode of harmonic generation. After setting the fundamental frequency, harmonic number and amplitude of each harmonic, we connect the circuit to the instrument^[8]. When the power is turned on, the APP automatically displays the distortion THD_x measurements of the current signal, normalizes the amplitude and restores a periodic waveform after the Bluetooth connection between the screen and the mobile phone. The accuracy of the observation device is measured by setting different frequencies and distortions.

If the nominal value of the signal source is THD_0 , the

THD absolute error calculation formula is:

$$THD_{error} = \frac{|THD_x - THD_0|}{THD_0} \times 100\% \quad (5)$$

Table 1 Signal Distortion and Harmonic Normalized Amplitude Test Data

Frequency (KHz)	THD0 (Nominal)	THDx (Measure)	Base wave Amplitude (mVPP)	THD Absolute Error
1	17.3%	17.6%	30	0.3%
1	22.4%	22.6%	600	0.2%
10	22.4%	22.8%	30	0.4%
30	28.3%	29.0%	30	0.7%
30	10%	10.5%	300	0.5%
50	36.1%	37.1%	100	1.0%
70	14.1%	15.6%	300	1.5%
70	44.7%	46.1%	600	1.4%
100	10%	9.9%	30	0.1%
100	44.7%	46.5%	300	1.8%
120	10%	10.8%	30	0.8%
120	44.7%	45.4%	450	0.7%
150	10%	11.1%	30	1.1%
150	44.7%	46.2%	600	1.5%

This system has completed the design and manufacture of a signal distortion measurement device based on C2000 MCU. The system takes TMS320F28335 as the main control MCU. After adjusting the input signal, ADC sampling and FFT operation are carried out to calculate the

harmonic component and total harmonic distortion THD_x , and normalized amplitude. The master MCU transmits the signal eigenvalues to TMS320F2812 to restore a periodic waveform and display it. The system measurements and screen displays refresh very quickly. The whole system is fully functional and the measurement results are very accurate. The whole process is automatically measured without manually adjusting the system parameters according to the frequency. The base frequency of distortion measurement reaches 150 kHz, and the system performance is stable. The displayed data is displayed on the mobile APP through Bluetooth transmission, and the human-computer interaction is friendly.

In summary, this new type of signal distortion analyzer has great superiority over the traditional one. The advantages of high precision and short-time measurement will have a profound impact on many products in the modern wireless communication process.

5 References

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